

PRELIMINARY · FOR CLIENT REVIEW



SMART-CONTRACT SECURITY ASSESSMENT · PRELIMINARY

Presale

HOODPAD Launchpad

Contract: Presale

Address: 0x9fBb8028...48722c

Chain: Robinhood Chain (4663)

Client: HOODPAD · voltipad.com

Prepared by: CFG Ninja

Date: September 6, 2026

Status: Preliminary (pending client response)

This is a **PRELIMINARY** report issued for client review. Findings and scores are subject to change after client response and a verification (re-audit) pass. Not for public distribution.

Overview

CONTRACT	Presale – Per-sale presale escrow, finalize & liquidity
ADDRESS	0x9fBb80283901B25E6BDacC0698a9Be4E0a48722c
CHAIN / COMPILER	Robinhood Chain (4663) · Solidity 0.8.34
CODEBASE	Flattened HOODPAD.sol (Factory + Presale + TaxToken). Every sale deploys fresh copies of the same code.

The presale handles contributions, refunds, finalization and liquidity provisioning. The token side of the pad is trust-minimized, but the PRESALE side is trust-heavy: on a successful sale the owner receives 85% of the raise up front and liquidity is provisioned only at the owner's discretion. Two high-severity centralization/correctness issues and one medium were identified. Buyers of any sale on this pad should understand these controls.

Result

84
/ 100

AUDIT PASSED

PASS – elevated centralization / rug risk

Score = 100 - (Critical×10 + High×5 + Medium×3 + Low×1). Pass threshold 80. Preliminary.

Vulnerability summary

SEVERITY	FOUND	PENDING	RESOLVED
Critical	0	0	0
High	2	2	0
Medium	1	1	0
Low	3	3	0
Total	6	6	0

All findings are Detected/Pending in this preliminary issuance; the Resolved column is populated in the final report after client remediation and re-audit.

Detailed findings

HOOD-01 | `abort()` after `finalize()` strands buyer refunds.

CATEGORY	SEVERITY	CVSS	LOCATION	STATUS
Access Control / Funds	High	7.5	Presale.finalize / abort / refund	Detected

Description. `finalize()` (callable by anyone once the sale ends) pays the owner 85% of the raise and keeps only 15%. The owner can then call `abort()`, which sets `softCapReached=false` and burns the held tokens. Buyers now fall into the `refund()` path (which requires `finalized && !softCapReached`), but the contract holds only ~15% of the raise – so refunds drain that and then revert, and `claim()` also reverts. Buyers of a SUCCESSFUL sale can lose ~85% of their contribution.

Recommendation. Disallow `abort()` once `finalize()` has released owner funds (e.g., block `abort` when `finalized && softCapReached`), or escrow the owner payout until liquidity is added / the refund window closes. Make the `finalize/abort` state machine mutually exclusive.

HOOD-02 | Liquidity is not guaranteed; 85% released to owner up front.

CATEGORY	SEVERITY	CVSS	LOCATION	STATUS
Centralization	High	7.1	Presale.finalize / graduate	Detected

Description. On a successful sale `finalize()` immediately sends 85% of the raise to the owner with no lock or vesting. Liquidity is added ONLY if the owner voluntarily calls `graduate()` (`onlyOwner`); nothing forces it. An owner can take the raise and never provide liquidity, leaving buyers holding claimable-but-illiquid tokens.

Recommendation. Automatically provision liquidity within `finalize()` (atomic), and/or vest or escrow the owner payout behind successful liquidity provisioning. Remove owner discretion over whether liquidity is ever created.

HOOD-03 | Systemic 15% liquidity ratio.

CATEGORY	SEVERITY	CVSS	LOCATION	STATUS
Tokenomics	Medium	5.0	Presale.LIQUIDITY_BPS	Detected

Description. `LIQUIDITY_BPS = 1500` is a hardcoded platform default applied to EVERY sale on the pad: 85% of the raise goes to the owner and only 15% to liquidity. Buyers cannot choose a higher lock ratio per sale. This is a high rug-risk ratio.

Recommendation. Make the liquidity ratio a per-sale, buyer-visible parameter with a sane minimum, or raise the default. Surface it prominently to contributors.

HOOD-04 | 20% early-withdraw fee.

CATEGORY	SEVERITY	CVSS	LOCATION	STATUS
Tokenomics	Low	2.0	Presale.EARLY_FEE_BPS	Detected

Description. `EARLY_FEE_BPS = 2000` charges a 20% fee to contributors who exit early via `withdrawEarly()`; the fee is paid to the launchpad.

Recommendation. Disclosed constant – ensure the 20% early-exit penalty is clearly communicated to contributors in the UI.

HOOD-05 | Duplicated finalize logic in graduate().

CATEGORY	SEVERITY	CVSS	LOCATION	STATUS
Code Quality	Low	1.5	Presale.graduate	Detected

Description. graduate() re-implements the finalize() success branch. Duplicated logic can diverge during maintenance and is harder to reason about.

Recommendation. Extract the shared finalize logic into one internal function called by both finalize() and graduate().

HOOD-06 | Standalone deployment trusts an arbitrary token.

CATEGORY	SEVERITY	CVSS	LOCATION	STATUS
Input Validation	Low	2.0	Presale.constructor	Detected

Description. Presale accepts an arbitrary token address in its constructor. Within the factory flow the token is always a freshly deployed TaxToken, but a Presale deployed directly (outside the factory) could be pointed at a malicious token.

Recommendation. For the pad, only use presales created by the audited factory. Optionally validate the token against the factory registry.

Positive observations

- Checks-Effects-Interactions followed in every state-changing path – no reentrancy found (state updated before external calls).
- Provisioned LP tokens are burned to the DEAD address = permanently locked liquidity.
- refundAll() is paginated (start/limit) – no unbounded-loop DoS.
- Per-address cumulative maxBuy enforcement; hardcap and window checks.
- Factory supply check guarantees enough tokens for claims + listing.

Scope & methodology

Manual line-by-line review of the flattened HOODPAD.sol as verified on Robinhood Chain, focused on access control, fund flows, reentrancy, tokenomics/tax logic, presale accounting, liquidity provisioning and centralization. On-chain bytecode presence and verification were confirmed via RPC and the block explorer. This preliminary assessment reviews the Presale logic; the launchpad deploys fresh copies of this exact code for every sale.

Disclaimer. This PRELIMINARY report is an engineering work product prepared by CFG Ninja for the client’s review. It reflects the code as reviewed on the date shown and is not final, not legal or financial advice, and not an endorsement or guarantee against loss. Severities and scores may change after client response and verification. Smart contracts carry inherent risk; deploy and invest at your own discretion.